

CS2108 - Introduction to Media Computing
Audit
Summary

Sim Ray En Ryan

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1 Introduction (L1)

1.1 Aims and Objectives

- This module introduces students to the concepts, issues, design, implementation, standards and applications of multimedia technologies.
- Special emphasis is placed on media representation, standards, content analysis and search.
- The module is divided into 2 parts:
 1. Fundamentals and Standards of Digital Media.
 2. Text, Image and Audio Search.

1.2 Fundamentals & Standards of Digital Media

- The most important development in media processing is the standardization of audio, image and video standards, such as MP3, JPEG and MPEG.
- These standards promote the wide spread use of audios, images, and videos on the Web.
- Students must have a fundamental understanding of the compression schemes, and their strengths and limitations.
- Topics covered include:
 - Fundamentals of digital media.
 - Compression techniques.
 - Audio standards.
 - JPEG, MPEG (for movies) and H26x (for video conferencing).

1.3 Text, Audio and Image Search

- Search is a fundamental tool for information access and management (Web, Social Networks, local file system).
- Media search is becoming active due to:
 - Exponential growth of social media contents.
 - Advances in media search, distributed computing and cloud technologies.
- Issues to be discussed:
 - Representation of text, audio and image content.

- Similarity match between a query and database items.
- Bridging of semantic and user intention-gaps in retrieval.
- Backend architecture.
- Search engine design and implementation.

1.4 Development of Search Engines

1.4.1 Scale and Content

- The web is really large (100 B pages) with lots of dynamically generated content.
- The biggest addition of info is social network sites.
- Current Web is text-based and document-oriented; the future is going to be multi-media, info and AI oriented.

1.4.2 Types of Search Engines

- **Conventional (Library Catalog):** Search by keyword, title, author, etc.
- **Text-based (Google, Bing, Baidu):** Search by keywords, limited natural language queries.
- **Multimedia (QBIC, WebSeek, Major Search Engines):** Search by visual appearance (shapes, colors, ...).
- **Question Answering Systems:** Search in (restricted) natural language.

1.4.3 Building a Search Engine

- **Core Steps:** Decide what to index (data/features), gather data (crawl), index it (efficiently), keep the index up to date, provide user-friendly query facilities.
- **Other Considerations:** Understand web structure for efficient crawling, understand user information needs, preprocess text and other unstructured data, cluster/-classify data, evaluate performance.

1.5 History of Media Search Engines

1.5.1 Early Media Search Systems

- Use of media contents in search evolved in the early '90s.
- **QBIC (IBM):** First complete commercial Content-Based Image Retrieval (CBIR) System.
 - Uses color, texture, shape features.

- Supports search by colors, shapes, sketch, and text (primitive).

1.5.2 CBIR in Main Stream Search Engines

- More recently, content-based media search has been integrated.
- Uses a combination of text and content-based media search functions.
- Media features: color, texture, visual keywords (SIFT), and deep learning features.
- Media features and text play complementary roles in retrieval.
- **Reasons for Recent Integration:**
 - Visual features are sufficiently robust (local visual features, deep learning features).
 - Large amount of social images with partial annotations are available.
 - Advances in indexing strategy and storage permit large visual index to be searched efficiently.

1.6 Applications of Media on Social Networks

- **E-Commerce Vertical Search:** Applications in fashion, home furnishing, schematics, logos, jewelry.
 - **ViSenze (NUS startup):** Specializes in visual fashion search (*snap-search-buy*). Business model is Enterprise Search, Content to Commerce, Content to Advertisement, and Visual Discovery.
- **Reverse Search Engine (TinEye):** Finds where an image came from and how it's being used. Technology is copy and near duplicate detection.
- **Photo Sharing (Instagram):** Use Case: Photo/video transform & sharing, Social networking. Model: Advertising.
- **Amazon Firefly:** Specializes in logo recognition and matching in real-time.
- **Snapchat:** Uses brand filters (geofilters) to reach younger audiences. Content flow: Stories → Live → Discover.
- **Trends:** Leveraging huge amounts of social images, increasingly requiring content-based matching capability, focused on real-time, social, and location-based features.

1.7 Course Arrangements (Summary)

- **Pre-requisite:** CS1020 (Object Oriented Programming).

- **Assessment:** Tutorial Participations (1%), Assignments (40%: 14+12+14), Midterm Test (24%-open book), Final Exam (35%-open book).
- **Assignments (Groups of 2):** 1) Image Search System, 2) Audio Matching System, 3) Open Project (Design of a media search app, preferably with proof-of-concept demo).

2 Introduction to Text Retrieval (L2)

2.1 Basic Concepts in Information Retrieval

- **Free Text:** Unstructured sequence of text units with uncontrolled vocabulary. Differs from DBMS (structured records).
- **Semi-Structured Text:** Contains structure (e.g., HTML tags, XML encoding).
- **Key Concepts:**
 - **Query Q:** Representation of what the user seeks.
 - **Document:** Entity to retrieve (document or text annotation of an image).
 - **Collection:** Set of documents.
 - **Index:** Representation of info that makes query easier.
 - **Term:** Word or concept in document or query.
- **Web Search:** Requires combination of content, link, and usage analysis.
- **User Query Characteristics:** Very short (typically 2 words), many typos, little use of advanced query operators.

2.2 Analysis of Text

- **Goal:** Extract the minimum set of distinct features to represent and distinguish a document.
- **Term Selection:**
 - **Stop-list:** Removes high frequency words (e.g., "the", "a") that are not useful for retrieval.
 - **Inverse Document Frequency (idf):** Terms frequent in a small number of documents are good for precision.

$$\text{idf} = \log_2 \left(\frac{N}{n_k} \right) + 1$$

(N : total docs, n_k : docs with term k).

- **tf.idf Weight:** Combines Term Frequency (tf , helps recall) and idf for a term k in document i .

$$w_{ik} = tf_{ik} \cdot \left[\log_2 \left(\frac{N}{n_k} \right) + 1 \right]$$

- **Stemming of Terms:** Reduces terms to their "morph" form to group variants of a word (e.g., RECOGNIZE, RECOGNIZED). Aims to reduce the number of terms and identify similar words.

2.3 Similarity Measures

2.3.1 Distance Measures

- **Euclidean Distance (L_2):** $Dist(\underline{D}_i, \underline{Q}) = \sqrt{\sum_{k=1}^N (d_k - q_k)^2}$.
- **City-block Distance (L_1 /Manhattan):** $CityB(\underline{D}_i, \underline{Q}) = \sum_{k=1}^N |d_k - q_k|$.

2.3.2 Similarity Measures

- **Jaccard Coefficient:** $Jac(\underline{D}_i, \underline{Q}) = \frac{|\underline{D}_i \cap \underline{Q}|}{|\underline{D}_i \cup \underline{Q}|}$ (more for sets).
- **Term Correlation:** $Corr(\underline{D}_i, \underline{Q}) = \sum (d_{ik} \cdot q_k)$.
- **Cosine Similarity (Normalization of Correlation):**

$$CosSim(\underline{D}_i, \underline{Q}) = \frac{\sum (d_{ik} \cdot q_k)}{\sqrt{\sum (d_{ik})^2 \sum (q_k)^2}}$$

- Cosine Similarity is widely used in high dimensional positive space and measures the angle between unit length vectors.

2.3.3 Vector Space Model (VS Model)

- Developed by Salton (SMART Retrieval Systems).
- **Assumptions:** Fixed size term set; terms are un-related (orthogonal).
- **Advantages:** Simplicity, handles weighted terms, easy to modify.
- **Disadvantages:** Term independent assumption, does not consider correlation (First order), synonym and polysemy problems.
- **Solutions:** Use thesaurus; apply mathematical analysis like SVD $\rightarrow$ LSA (Latent Semantic Analysis).

2.4 Inverted Index

- Instead of a term-document matrix, an inverted index (posting table) is used for storage efficiency on large collections.
- **Structure:** Maps terms to the list of documents containing them (e.g., *vector* $\rightarrow$ File1, File2, File3).
- **Advantage:** For a short query, it only requires a few file accesses, independent of the total number of documents (N).
- **Operations:** Conjunction (AND) is an iterative merge, Disjunction (OR) is similar, Negation (NOT) can also be done efficiently.

2.5 Relevance Feedback (RF)

- Uses users' judgments (relevant/non-relevant documents) to modify the search model.
- **Query-Space RF:**
 - **Idea:** Re-formulate the query based on relevant/non-relevant documents.
 - Terms from relevant documents are added (or weights increased); terms from non-relevant documents are deleted (or weights reduced).
 - **Formula:** $\underline{Q}^{(i+1)} = \underline{Q}^{(i)} + \alpha \cdot \sum_{D_i \in R} \underline{D}_i - \beta \cdot \sum_{D_j \in NR} \underline{D}_j$
 - **Pseudo RF:** Automates the manual part by assuming the top k documents are relevant.
 - **Limitation:** Benefits only the current session (not long term) and does not use the wisdom of other users.
- **Document-Space RF (DRF):** Modifies the documents (e.g., adding text annotation to images) to make them more easily retrievable in the future by other users.

2.6 Evaluation Measures

- **Metrics (based on a query):**
 - **Precision (P):** Fraction of retrieved documents that are relevant. $P = a/(a + b)$.
 - **Recall (R):** Fraction of relevant documents that are retrieved. $R = a/(a + c)$. (a : Retrieved & Relevant, b : Retrieved & Non-Relevant, c : Missed & Relevant).
- **Single Value Measures:**
 - **Average Precision (AveP):** Average of precision values computed at the rank of every relevant document.

- **Mean Average Precision (MAP)**: Average of AveP over all queries: $MAP = \frac{\sum_{q=1}^Q AveP(q)}{Q}$.
- **F_1 Measure**: Harmonic mean of Precision and Recall (when $\beta = 1$): $F_1 = \frac{2 \cdot P \cdot R}{P + R}$.

3 Introduction to Content-based Image Retrieval (L3)

3.1 Color-Based Image Retrieval

- Computers see images as a matrix of pixel values; the simplest representation is the color histogram.
- **Color Histogram**: Collects the frequency of color values in different quantization levels/bins.
- **Properties**: Fast to compute, small dimension, and correlates well with image difference (monotonicity).
- **Limitations**: No spatial info, not robust to large appearance changes (e.g., rotation).

3.2 Similarity Measures and Matching (Image)

- **Color Representation**: Use perceptually linear color spaces (Luv, Lab, YUV, YC_rC_b) instead of RGB, as the difference between colors corresponds well with human perception.
- **Histogram Matching**: Simple correlation ($\underline{Q} \cdot \underline{D}_i$) is flawed as it does not consider similarity between colors.
- **Perceptual Similarity**: Use a pre-computed perceptual color similarity matrix $S(i, j)$ (similarity between colors i and j) to refine matching: $Sim_{per}(\underline{Q}, \underline{D}_i) = \underline{Q}^T S \underline{D}_i$.
- **Feature Fusion**: When combining multiple features for similarity, a weighted sum is used: $Sim(\underline{Q}, \underline{D}_i) = \sum_j^m \alpha_j Sim(\underline{Q}^j, \underline{D}_i^j)$, where $\sum_j \alpha_j = 1$.

3.3 Other Color and Texture Representations

3.3.1 Color Representations

- **Segmented Image Representation**: Divides the image into sub-windows (e.g., 4×4) and computes a histogram for each, partially capturing local information.
- **Color Coherence Vector (CCV)**: Forms two histograms, one for colors in large regions (H_α) and one for isolated colors (H_β), with higher weight given to H_α .

- **Color Moment:** Represents color content using moments (Mean, Variance). Simple and efficient, but lacks descriptive power.

3.3.2 Texture Representation

- **Texture:** Something that repeats with variation.
- **Tamura Features:** Classify textures based on: Coarseness, Contrast, Directionality, Linelikeness, Regularity, and Roughness.
- **Edge Representation (Histogram):** Converts image to luminance Y , uses a Sobel operator to find edge magnitude D and gradient ϕ . Texture is represented as an Edge Histogram $H(\phi)$ (e.g., 8 directions).

3.4 kNN-based Image Search and Classification

- **CBIR Goal:** Given an image query, extract features, perform similarity matching, and return the top k matches (k-Nearest Neighbors).
- **kNN Classification:** Given a new image, perform CBIR to retrieve the top k labeled images and use their labels for a majority vote on the new image's label.
- **Key Lesson:** At large scale (100s of million to billion size), simple robust methods (like kNN) work best, especially with advanced visual features.

4 Advanced Visual Features and Multimedia Indexing (L4)

4.1 Intro to Advanced Visual Features

- **Local Interest Points (LIPs):** Used for object-level matching.
- **Good LIP Characteristics:** Highly distinctive and repeatable; Invariant to uniform scaling, rotation, changes in illumination, and viewpoint changes.
- **Detection:** Corner detection (Harris, rotation invariant) or Scale Invariant Blob Detection (Laplacian-of-Gaussian).

4.2 Scale Invariant Feature Transform (SIFT)

- **Descriptor Idea:** Use edge orientation representation around the detected feature point.
- **Mechanism:** Divides a 16×16 window into a 4×4 grid (16 cells) and computes an 8-orientation histogram for each cell, resulting in a $16 \times 8 = 128$ dimensional descriptor.

- **Invariance:** Invariant to Scale and Rotation; partially invariant to illumination and viewpoint changes.

4.3 Visual Keywords (Bag of Visual Words)

- **Need:** Feature descriptors have too many variations and must be quantized to a smaller, fixed number of "visual keywords."
- **Process (Quantization):**
 1. Extract and collect SIFT descriptors from training images.
 2. Cluster features (e.g., using Hierarchical K-means) to create a discrete codebook of visual words.
 3. Represent each image as a histogram of visual words.
- **Limitation:** Image queries contain thousands of visual words, unlike short text queries, leading to high computation costs.

4.4 Visual Concept Detector

- **Concept Recognition** involves: Verification (Is that X?), Detection (Are there X?), Identification (Is that X, a landmark?), Object Categorization, and Scene/Context Categorization.
- **Semantic Concept Features:** Tools output a high-dimensional vector (e.g., 1000-D) where each element is the possibility of the image belonging to a category.
- **Tools:** 1000-Class Image Classification Tool (HOG, SIFT, LBP, SVM); BVLC CaffeNet (Deep Learning); Clarifai's Deep Learning System.

4.5 Intro to Multimedia Indexing

- **Need:** Linear Scan (comparing query to all images) is too slow for large-scale databases. Indexing pre-computes an index to improve search speed.
- **Indexing Methods:** Inverted File, Hashing-based Indexing, KD-Tree.

4.6 MM Indexing Methods

4.6.1 Inverted File Indexing (for Visual Words)

- Builds an inverted index mapping visual words to image numbers.
- Search involves extracting visual words in the query and accessing the index to find intersecting lists of matching images.

4.6.2 Hashing-based Indexing

- **Idea:** Hash high-dimensional data into a low-dimensional Hamming space so that similar samples map to the same (or adjacent) hash buckets.
- **Hamming Distance:** Number of different bits between two binary strings.
- **Locality Sensitive Hashing (LSH):** Popular approach based on random projection.

– **Hashing Function:**
$$h_r(x) = \begin{cases} 1, & \text{if } r^T x \geq 0 \\ 0, & \text{otherwise} \end{cases}, \text{ where } r^T x \geq 0 \text{ is a hyperplane.}$$

4.6.3 KD-Tree

- A binary tree storing k -dimension data samples by recursively partitioning the space.
- Effective for low-dimensional data (< 10 dimensions), but degrades to near linear scan for high-dimensional data.

4.7 MM Indexing Strategy

- Different features use different indexing structures:
 - **KD-Tree:** Global dense features (color/texture histograms).
 - **Inverted Structure:** Bag-of-Visual-Words Feature.
 - **Hashing Index:** Global and combined (fused) features.

5 Fundamentals of Digital Media (L5)

5.1 Digital vs. Analog Signals

- **Analog Signal:** Can vary continuously over space/time, represented as $f(x, y, z, t)$ (continuous values at continuous space/time).
- **Digital Signal:** Discrete values at discrete space/time.
- **Waves:** Energy propagation characterized by:
 - **Amplitude:** Intensity/loudness/brightness.
 - **Frequency:** Pitch/color.
- J.B. Fourier demonstrated that any irregular wave can be constructed from a combination of sine & cosine waves.

5.2 Analog to Digital (A-D) Conversion

1. **Sampling:** Records wave values at pre-determined discrete points/intervals (time or space discretization).
2. **Quantization:** Maps continuous values to a finite set of discrete values (quantization levels).

5.2.1 Sampling Theory

- **Nyquist's Theorem:** The sampling frequency must be greater than twice the maximum frequency of the input signal to accurately reconstruct the original signal.
- **Under-sampling (Aliasing):** Occurs when sampling frequency is less than the Nyquist rate, failing to reproduce the cycles of the original wave.

5.2.2 Quantization

- **Quantization Error:** Distortion caused by representing a continuous value with a discrete level.
- **Resolution:** Intensity resolution (L) depends on the number of bits (p): $L = 2^p$.
- **Total Storage:** $b = M \times N \times p$ bits ($M \times N$ is image size, p is bits/pixel).

5.3 Digital Image Representation

- **Effects of Low Resolution:**
 - **Spatial Resolution:** Checkerboard Effect (resolution decreased too much).
 - **Intensity Resolution:** False Contouring (not enough gray scales in smooth areas).
- **Color Image (24-bit):** Modeled by $\{R, G, B\}$ triplets (8 bits per channel, ~ 16.7 million colors).
- **Color Compression (8-bit/256 colors):**
 - **Color Look-up Table (CLUT):** Image stores an index (e.g., 8-bit) that points to a 24-bit color in the lookup table, achieving large storage savings.
 - **Median-cut algorithm:** Used to select the best 256 colors by recursively splitting the widest color range at the median.

5.4 Fourier Transforms

- **Function:** Transforms a signal f from the spatial domain to the frequency domain F (and back) without loss.

- **Why Frequency Domain?:** Operations like filtering (low-pass, high-pass) and compression (removing high-frequency components) are easier.
- **2D Discrete Fourier Transform (DFT):**

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

- **Properties:** Invertibility, Separability, Rotation invariance.
- **Convolution Theory:** Convolution in the spatial domain is equivalent to multiplication in the frequency domain: $f(x) * g(x) \leftrightarrow F(u)G(u)$.
- **Block Transforms (DCT):** Discrete Cosine Transform (DCT) operates on blocks (e.g., 8×8). It is widely adopted (JPEG/MPEG) as it approximates the optimum Karhunen-Loeve Transform (KLT).

6 Introduction to Digital Audio (L6)

6.1 Introduction to Audio

- **Sound:** Energy from vibrating objects, propagating via compressions and rarefactions.
- **Characteristics:**
 - **Loudness:** Physical manifestation of amplitude (measured in decibel, dB).
 - **Pitch:** Physical manifestation of frequency (measured in Hertz, Hz). Human range: 20 Hz to 20 KHz.
 - **Timbre/Quality:** Physical manifestation of waveform shape.

6.2 Psycho-Acoustics

- **Decibel (dB):** Unit for measuring perceived loudness (logarithmic ratio of power/amplitude).
- **Minimum Hearing Threshold:** Least audible sound (max sensitivity at 1 – 5 KHz).
- **Amplitude Masking:** An audible sound shifts the hearing threshold up in its frequency region (critical band), making quieter sounds inaudible.
- **Temporal Masking:** After a loud sound, there is a delay before the ears can hear a quieter sound.

6.3 Digital Audio

- **Nyquist Sampling Theorem:** The sampling rate must be > 40 KHz for the human hearing range (20 Hz to 20 KHz).
- **Signal to Quantization Noise Ratio (SQNR):** $SQNR = 6.02N$ (dB), where N is the number of bits per sample.
- **Uncompressed Data Rates:** Stereo CD quality: 44.1 KHz sampling, 16 bits/sample, ~ 176.4 KB/sec.

6.4 Audio Coding and Compression

- **Pulse Code Modulation (PCM):** Basic sampling and quantization (results in high data rates).
- **Differential PCM (DPCM):** Codes the *difference* (prediction error) between the actual sample value and a predicted value (derived from previous samples). Uses fewer bits as differences are smaller.
- **Delta Modulation (DM):** Special case of DPCM where difference is coded with a single bit. Prone to Slope Overload Distortion (signal too fast) and Granular Distortion (signal too slow).
- **Adaptive DPCM:** Varies the number of bits/quantized step size based on the differential signal amplitude to minimize distortion.

6.5 Audio Analysis and Features

- **Perceptually Relevant Features:** Loudness, Pitch, Tone, Mel-filtered Cepstral Coefficients (MFCC).
- **Basic Features:** Zero crossing rate, short time energy, MFCC.
- **MFCCs (Mel-frequency cepstral coefficients):** A representation of the short-term power spectrum based on a linear cosine transform of a log power spectrum on a non-linear *mel scale* of frequency. The mel scale approximates the human auditory system's response better than linear frequency.

6.6 Audio Standards

- **WAV / AIFF:** Uncompressed/Lossless formats.
- **MPEG-1 Audio Layer 3 (MP3):** Target rate ~ 128 kb/s for stereo. CD-quality over low bit rate channels.
- **MPEG-2 AAC (Advanced Audio Coding):** Not backward compatible with MPEG-1, codes at 64 kbps/channel.

7 Image Filters & Transformation (L7)

7.1 Dithering

- **Definition:** Simulates unavailable colors by blending available colors to fool the eye (trades intensity resolution for spatial resolution).
- **Techniques:**
 - **Half-Tone Printing:** Replaces a pixel with a dot pattern.
 - **Ordered Dithering:** Uses an array (Dither matrix) on pixel blocks.
 - **Floyd-Steinberg Dithering:** Propagates the quantization error to neighboring unprocessed pixels (achieves a better visual quality).

7.2 Image Filters (Point Processing)

- * **Point Operations:** Modify pixels independently ($y = f(x)$). Used for contrast adjustment.
- * **Examples:** Identity, Negate ($f(x) = 1 - x$), Brighten/Darken ($f(x) = x^p$).
- * **Tone Correction:** Adjusting RGB components equally preserves hues. High-key/Low-key images have info concentrated at high/low intensities.

7.3 Image Filters (Spatial Filtering)

- * **Convolution:** Output pixel is the sum of products of the input image $a(u, v)$ and a filter mask $h(x - u, y - v)$.

$$b(x, y) = \sum_{u=-\infty}^{+\infty} \sum_{v=-\infty}^{+\infty} a(u, v) \cdot h(x - u, y - v)$$

- * **Examples of Filters:**
 - **Blurring:** Averaging mask (used for anti-aliasing).
 - **Median Filter:** Non-linear filter for noise reduction (eliminates spikes).
 - **Edge Detection (Sobel):** Approximates gradient to discover edges.

7.4 Color Transfer between Images

- * **Goal:** Scale and shift one image's color distribution to resemble another image's.
- * **Method:**

1. Convert images to a perceptually linear, non-correlated color space (e.g., $L\alpha\beta$).
2. Compute mean (μ) and standard deviation (σ) for each color axis.
3. Apply linear transform (shift and scale) to the source image (C_s):

$$C' = \frac{\sigma_t}{\sigma_s}(C_s - \mu_s) + \mu_t$$

(σ_t, μ_t : target image statistics; σ_s, μ_s : source image statistics).

8 Lossless Compression Techniques and Monochrome JPEG (L8 & L9)

8.1 Digital Compression

- * **Need:** Reduce storage requirements and data transfer time.
- * **Principle:** Find and reduce redundant information.
- * **Redundancy Types:**
 - **Statistical:** Exploited by entropy coding (e.g., Huffman, RLC).
 - **Psycho-visual:** Exploited by quantizing less important information (Lossy Compression).
- * **Types of Compression:**
 - **Lossless:** Original data recoverable; low compression ratio ($2 - 5\times$).
 - **Lossy:** Data is an approximation; high compression ratio ($> 10 - 100\times$).

8.2 Lossless Compression Techniques

- * **Entropy (η):** Minimum # of bits required to encode symbols; compression is possible when probability is skewed ($\eta \leq \bar{l} \leq \eta + 1$).
- * **Run-Length Coding (RLC):** Replaces sequences of identical bytes with the byte and count (e.g., $BBBBBBBBB \rightarrow B!9$).
- * **Huffman Coding:** Assigns shorter codewords to more frequent symbols (Unique prefix property, optimal redundancy code).
- * **Arithmetic Coding:** Encodes a message sequence by an interval defined by its probability, overcoming the ≥ 1 bit per symbol limitation of Huffman coding.

- * **LZW Coding (Lempel-Ziv-Welch):** Dictionary-based. Uses fixed-length codewords for variable-length strings, building the dictionary adaptively on the fly (used in GIF).
- * **Predictive Encoding:** Lossless, pixel-level scheme using a linear predictor (e.g., $X = A + B - C$) to estimate the next pixel. Encodes the difference (Diff) and the predictor code.

8.3 Lossy Compression Techniques

- * **Quantization:** Main source of compression and loss. Reduces distinct output values.
- * **Transform Coding (DCT):** Transforms signal from spatial domain to frequency domain ($\underline{Y} = T(\underline{X})$) to de-correlate input and concentrate energy into a few coefficients.

8.4 Monochrome JPEG Standard

8.4.1 JPEG Compression Phases

1. **Image Preparation:** Divide image into 8×8 blocks; shift pixel values to be centered about zero (e.g., $0 - 255 \rightarrow -128$ to 127).
2. **Image Processing:** Apply 2D DCT to each 8×8 block, yielding 1 DC and 63 AC coefficients.
3. **Quantization:** Divide each DCT coefficient $F(u, v)$ by a corresponding value $Q(u, v)$ from a Quantization Table. This is the **major compression step**.
4. **Entropy Encoding:** Order the resulting matrix in a zig-zag sequence. Encode DC coefficients as differences from the previous block. Use RLE of zero values followed by Huffman/Arithmetic coding for all coefficients.

8.4.2 Image Decoding

1. Decode Entropy coded data.
2. Dequantize: $F^R(u, v) = F^Q(u, v) \times Q(u, v)$.
3. Apply Inverse Discrete Cosine Transform (IDCT) to convert back to pixel domain for display.

9 Color Model and Color JPEG (L10)

9.1 The Color Models

- * **Perception:** Color is perceived by Hue, Saturation, and Brightness (Chromaticity = Hue + Saturation).
- * **Hardware Models:**
 - **RGB:** Additive, used for displays (non-linear).
 - **CMY(K):** Subtractive, used for printing.
- * **Broadcast/Compression Models:**
 - YC_rC_b : Used for digital image/video compression (JPEG/MPEG). Y =Luminance, C_r, C_b =Chrominance.
- * **Human-Oriented Models:** HSI, HSL, HSV, L^*a^*b (perceptually uniform, orthogonal axes).

9.2 Color JPEG Standard

- * **Color Space:** Adopts the YC_rC_b model.
- * **Compression through Sub-sampling:** Chrominance components ($\{C_r, C_b\}$) are often reduced 2 : 1 horizontally (e.g., 2h1v sampling), exploiting the eye's lower sensitivity to color than to luminance (Y). This provides initial compression (1/3 data saving).
- * **Quantization (Color):** Separate quantization tables are used for luminance and chrominance (larger quantization values used for higher-frequency AC coefficients, especially in chrominance).
- * **Compression Ratio:** A scale factor S can be used to scale the quantization table $Q(u, v)$ to adjust the overall compression ratio.

9.3 JPEG Modes

- * **Sequential Mode (Default):** Encodes the image in a single top-to-bottom scan.
- * **Progressive Mode:** Delivers low-quality versions quickly, improving quality over successive passes:
 - **Spectral selection:** Encodes low-frequency components (DC/first few AC) first.

- **Successive approximation:** Encodes Most Significant Bits (MSBs) first.
- * **Hierarchical Mode:** Encodes at different resolutions. Uses down-sampling and then encodes the difference between the actual and up-sampled lower-resolution image.
- * **JPEG 2000:** New standard using Discrete Wavelet Transform (DWT), offering superior quality/compression (no blocking artifacts), and supporting larger images.

10 Case Studies of MM Apps (L10)

10.1 Clarifai (Image/Video Recognition)

- * An AI company specializing in visual recognition (since 2013).
- * Technology built on advanced machine learning systems (Deep Learning APIs for object recognition).
- * Business Model: Tiered API pricing plans (B2B model).

10.2 ViSenze (Transforming Commerce through AI)

- * Specializes in visual commerce solutions.
- * Products include Mobile Visual Search, Product Recommendation, Price Comparison, and Catalogue Enrichment.
- * Business Model: Enterprise-facing (B2B) providing APIs for Content to Commerce/Advertisement and Visual Discovery.

10.3 Olapic (Discovering Actionable Contents)

- * Earned Content Platform to amplify engagement with curated, high-converting images and videos taken by real people.
- * **Process:** Curate (collect content), Activate (request rights, tag products, power marketing/e-commerce strategies), Analyze (content analytics on revenue/performance/engagement).
- * Business Model: B2B.

11 Digital Video and MPEG Standards (L11)

11.1 Introduction to Digital Video

- * **Motion Video:** A combination of image and audio, consisting of a sequence of still images called frames.
- * **Frame Rate:** Frames per second (fps). Should be ≈ 30 fps for smooth, realistic motion perception.
- * Audio is added and synchronized with the apparent movement of images.
- * **Advantages of Digital Video:**
 1. Can be stored on digital devices/memory, ready for processing.
 2. Direct access is possible, enabling simple nonlinear video editing.
 3. Repeated copying does not degrade image quality.
 4. Ease of encryption and better tolerance to channel noise.

11.2 Color Model for Digital Video

- * **Human Color Perception:** Based on rods (brightness/grey) and three groups of cones (blue, red, green).
- * **Eye Sensitivity:** Greatest for green-yellow range, decreasing towards red and blue ends of the spectrum.
- * **Digital Video Format:** Adopts the YC_rC_b format (Luminance Y , Chrominance C_r, C_b).
 - $Y = 0.3R + 0.59G + 0.11B$ (Reference white: 30% red, 59% green, 11% blue).
 - $C_b = B - Y$
 - $C_r = R - Y$
- * **Decimation:** Since humans see color with much less spatial resolution than luminance, it makes sense to "decimate" (sub-sample) the chrominance signal.

11.3 Video Representation: Chroma Sub-sampling

- * Various schemes for chroma sub-sampling:
 - **4:4:4:** Implies no chroma subsampling (full Y, C_b, C_r info).

- **4:2:2:** For every 4 units (pixels) of Y info, 2 units each of C_b and C_r are transmitted.
- **4:1:1:** The chroma is sub-sampled at one-fourth the horizontal frequency as luma.
- **4:2:0:** Performs both horizontal & vertical sub-sampling. For every 4 pixels of Y information, there are 2 pixels of C info along a row and along a column.

11.4 Temporal Redundancy and Motion Compensation

11.4.1 Video Compression Bottleneck

- * Raw data rate for CD quality audio is ≈ 700 kbps.
- * Raw data rate for High Definition Video is ≈ 663.5 Mbps, approximately 1,000 times the audio size.
- * Compression, especially temporal redundancy removal, is KEY.

11.4.2 Types of Redundancy in Video

- * **Spatial redundancy:** Reduced by intra-frame coding (e.g., JPEG on a single frame).
- * **Perceptual redundancy:** Reduced by quantization and differential coding of color components.
- * **Coding redundancy:** Reduced by Run-length and Huffman coding.
- * **Temporal redundancy:** Consecutive frames carry similar contents. Removed by inter-frame coding, resulting in the most significant reduction.

11.4.3 Principle of Temporal Redundancy Removal

- * Divide frame content into non-overlapping blocks.
- * Find identical or similar blocks in previous or subsequent frames.
- * If an identical block is found, need only to code the Motion Vector: $\{dx, dy\}$ (maximum saving).
- * Otherwise, code the difference between the current and predicted block: $\{dx, dy, \delta B\}$, where $\delta B = B_k(x, y) - B_{k-1}(x, y)$.
- * **Bi-directional search:** Considering a previous and a following frame (Interpolative Coding) offers better chance of finding a similar block and handles occlusion.

11.4.4 Block-based Motion Compensation Model

- * **Principle:** Predict contents of current frames (at block level) from previous or subsequent frames.
- * **MPEG Unit:** Macroblocks (MBs), typically 16×16 pixels (or 2×2 DCT blocks).
- * **Motion Information:** Comprises the direction and amplitude of displacement, $\{dx, dy, \delta B\}$.
- * **Advantages:** Low overhead, availability of low-cost VLSI implementation.
- * **Disadvantages:** Fails for zoom/rotation/local deformation; discontinuity at block boundaries (blocking artifacts).
- * **Aim (Block Translation Model):** Look for a best matched block in the search window of Frame $(k - 1)$ such that:

$$\text{Error} = \min_{\bar{v}} \sum_{\bar{x} \in \text{Block}} \text{Diff}[s_k(\bar{x}) - s_{k-1}(\bar{x} + \bar{v})]$$

11.5 Motion Vector Search Procedures

11.5.1 Matching Criteria for Macroblocks

- * **Minimum Mean Square Error (MSE):**

$$MSE(d_1, d_2) = \frac{1}{N_1 N_2} \sum_{(n_1, n_2) \in S} [s_k(n_1, n_2) - s_{k-1}(n_1 + v_1, n_2 + v_2)]^2$$

- * **Minimum Mean Absolute Error (MAD):**

$$MAD(d_1, d_2) = \frac{1}{N_1 N_2} \sum_{(n_1, n_2) \in N} |s_k(n_1, n_2) - s_{k-1}(n_1 + v_1, n_2 + v_2)|$$

11.5.2 Search Strategies

- * **Full Search:** Evaluates matching criterion at $(2p + 1) \times (2p + 1)$ distinct points for each block. Complexity: $O(p^2 N^2)$. Too complex.
- * **2D Logarithmic Search (3-Step Search/Cross Search):** Searches in selected locations (assuming motion changes slowly/smoothly) to minimize efforts. Complexity: $O(\log p \times N^2)$.
- * **Hierarchical Search (Multi-resolution):** Initial motion vector estimation obtained from significantly reduced resolution images, followed by refinement at higher resolutions. Offers the lowest computational cost.

11.5.3 Encoding Based on Search Results

- * Encode the motion vector $\underline{m}_v = (u, v)$ and the matching error $D(x_0, y_0)$.
- * **Skipped Macroblock:** If error $< \sigma_1$ (set $D = null$) and $\underline{m}_v = null$.
- * **Intra-coded Macroblock:** If error $> \sigma_3$ and $\underline{m}_v = null$ (no motion vector).
- * **Predicted Macroblock:** Otherwise, code the block using $D(x_0, y_0)$ and $\underline{m}_v$.
- * The error image $D(x_0, y_0)$ is coded as a normal JPEG image but uses a different quantization matrix (optimized for the flatter/zero-dominant residual error image).

11.6 MPEG Frame Types

- * **I-frame (Intra-coded Picture):** Coded entirely without reference to other frames (like a still JPEG image). Provides random access points and re-synchronization points.
- * **P-frame (Predictive-coded Picture):** Contents generated based on contents of a **previous** reference frame. Coded using motion vectors and error image.
- * **B-frame (Bi-directionally Predictive/Interpolative-coded Picture):** Contents generated based on both a **previous** and a **successive** reference frame. Offers better compression ratio, deals with uncovered areas properly, and allows decoupling between prediction and coding (no error propagation).

11.7 MPEG Standards (MPEG-1 and MPEG-2)

11.7.1 MPEG-1 (1992)

- * **Target Bit-rate:** Up to 1.5 Mb/sec (e.g., CD-ROM quality).
- * **Video Type:** Supports only non-interlaced video.
- * **Resolution:** 352×240 (NTSC) or 352×288 (PAL).
- * **Subsampling:** Uses 4:2:0 chroma subsampling.
- * **Application:** Utilized by VCDs.

11.7.2 MPEG-2 (1994)

- * **Target Bit-rate:** 2 – 15 Mb/s (optimized at 4 Mbps).

- * **Video Quality:** Provides video quality from NTSC/PAL up to HDTV quality.
- * **Compatibility:** Superset of MPEG-1 and backward compatible.
- * **Features:** Additional support for interlaced video, panning window within a frame, and encoding at multiple quality levels.
- * **Application:** Utilized by DVDs.
- * **Scalability:** Provides 4 levels (Low, Main, High 1440, High) and 5 profiles (Simple, Main, Spatial Resolution, Quantization Accuracy, High), all downward compatible, which permits partial implementation and provides for scalability.

11.7.3 MPEG Encoding Process

- * **Core Trade-off:** Balance random access (I-frame) with compression ratio (inter-frame: P/B frames).
- * **Macroblock Structure (MPEG-1):** A 16×16 area partitioned into 4 luminance (Y) blocks and 2 chrominance (U, V) blocks (4:2:0 sampling assumed).
- * **Group of Pictures (GOP):** The basic frame sequence organization (e.g., [IBBPBB] ...).
- * **Decoding Order (Example):** For a sequence I-B-B-P (frames 1, 2, 3, 4), the decoding order is 1 (I), 4 (P), 2 (B), 3 (B). P and B frames require preceding I/P frames to be decoded first.
- * **Impact of Frame Structure:**

Frame Pictures	Compression	Random Access	Coding Delay
I Pictures only	Low	Highest	Low
I and P Pictures	Medium	Medium	Medium
I, P and B Pictures	High	Low	High

- * **Bit Stream Syntax (6 Layers):** Sequence Layer $\rightarrow$ Group of Pictures Layer $\rightarrow$ Picture Layer $\rightarrow$ Slice Layer $\rightarrow$ Macroblock Layer $\rightarrow$ Block Layer.

11.8 Case Study: Micro-Videos and Classifieds (Review from L10/11)

- * **Vine (Micro-Video Sharing):** 6-second looping videos. Uses multimodal data (social, visual, acoustic, textual). Business Use: Marketing, Brand building (quick to digest, entertaining, measurable).

- * **Carousell (Mobile Classified App):** Mobile-first platform. "Snap to Sell, Chat to Buy." B2C (Consumer-facing).

12 H261 & Other Video Standards and Current Trends (L12)

12.1 Introduction to H.261

12.1.1 Need for Symmetric Video Standards

- * **MPEG:** Asymmetric (Encode once, Playback many times, e.g., VoD). Encoding need not be real-time.
- * **H.261:** Symmetric (Encode/Decode must be real-time, e.g., Video Conferencing).

12.1.2 H.261 Standard Details

- * Recommended standard for video phone, videoconferencing over ISDN lines.
- * **Bit-rate:** $p \times 64$ kbps, where $p \in \{1..30\}$ (also known as $p \cdot 64$ standard).
- * **Delay:** Encoder delay < 150 msec (required for real-time bi-directional conferencing).
- * **Frame Sizes** (using 4:2:0 sampling):
 - **CIF (Common Intermediate Format):** 352×288 luma with 176×144 chroma.
 - **QCIF (Quarter CIF):** 176×144 luma with 88×72 chroma.

12.1.3 Real-time Encoding in H.261

- * **No B-frames:** Removes complex encoding modes and high delay.
- * **P-frame Reference:** All P-frames reference only the nearest preceding I-frame, which permits parallel processing (Example sequence: $\{IPP \ PIPPPI \dots\}$).
- * **Motion Search:** Utilizes hierarchical search to speed up motion estimation.
- * **Rate Control:** Vary compression ratios (quantization step size) in real-time to ensure a constant encoding rate.
- * **Bit-stream Design:** Format is designed to facilitate skipping of data (e.g., entire GOBs) as necessary.

12.1.4 H.261 Basic Coding

- * **Basic Unit:** Macroblock (16×16).
- * **I-Frame:** Uses DCT, followed by a **Uniform Quantization** (as opposed to MPEG's table) for all coefficients, and then RLE & Huffman coding.
- * **P-Frame:** Encodes the motion-compensated difference (residual error) of the target frame with respect to the decoded previous I-frame.

12.2 Rate Control in H.261

- * **Problem:** Encoder output rate (depending on scene complexity) varies, but the channel rate (e.g., ISDN) is constant.
- * **Solution:** The encoded bitstream is buffered. Buffer fullness is fed back to the encoder (Rate Control mechanism).
 - **Active Scene (Buffer Fills Up):** Increase Quantization step size $\rightarrow$ higher compression/lower quality/lower encoder output rate.
 - **Static Scene (Buffer Empties):** Decrease Quantization step size $\rightarrow$ lower compression/higher quality/higher encoder output rate.
- * **Effect:** Quality varies considerably (static scenes $\rightarrow$ high quality, active scenes $\rightarrow$ lower quality) to maintain a constant bit rate.

12.3 H.261 BitStream Format

- * **Structure:** Picture $\rightarrow$ Groups of Blocks (GOBs) $\rightarrow$ Macroblocks (MB) $\rightarrow$ Blocks.
- * **GOB:** A block of 3×11 MBs, with a header to permit skipping.
- * **Header Information for Skipping:**
 - **PSC** (Picture Start Code), **TR** (Temporal Reference for timestamp), **PType** (Picture Type: I or P).
 - **GOB Start**, **Grp#** (Group Number), **GQuant** (Group Quantization Value).
 - **MB Addr** (Macroblock Address), **MB Type**, **Quant**, **M. Vector**, **CBP** (Coded Block Pattern, bitmask for present blocks).
- * The design allows for skipping entire pictures, GOBs, or macroblocks to maintain real-time constraints.

12.4 Other MPEG Standards

12.4.1 MPEG-4

- * **Aim:** Better compression and support for user interactivities (e.g., interactive TV).
- * **Key Feature: Object-based coding** (Video Object Plane, VOP).
- * Allows composition and content-based manipulation of individual VOPs.
- * **Note:** Object detection is **not** part of the standard. Deployment problems persist.

12.4.2 MPEG-7

- * **Aim:** Provide a description language (meta data style) to describe AV content for content-based retrieval applications (e.g., digital libraries).
- * **Main Elements:**
 - **Description Tools:** To define Descriptors (D) and Description Schemes (DS).
 - **DDL (Description Definition Language).**
- * **Examples of Descriptors:**
 - **Visual:** Color, Texture, Shape, Motion, Face Recognition, Spatial/Temporal object localization.
 - **Audio:** Sound effects, Musical Instrument Timbre, Spoken Content, Melodic Descriptors.

12.4.3 Future Trends in Standards

- * Standards up to MPEG-2 focused on compression; MPEG-4/7/21 have objectives like interactivity and search, which are harder to standardize and deploy.
- * New proposals look into incorporating user tags & Metadata (like EXIF) into JPEG/MPEG standards.

12.5 Current Trends: Social and Mobile Media Computing

- * **Context:** Wide-spread adoption of mobile devices and social networks leads to a proliferation of User-Generated Contents (UGCs).

- * **Key Trends:**
 - **Mobile Search:** Characterized by "snap and search" paradigm (real-time and location-based).
 - **Social Media Search:** Problems in information gathering; requires social analytics.
 - **Video Search:** The web is becoming more visual.
- * **Fashion Search (Street-to-Shop):** Cross-scenario retrieval via parts alignment and auxiliary sets. Key technologies include:
 - Visual Recognition (to segment people, clothing, and accessories).
 - Object Detection and Classification (e.g., using Google's TensorFlow, ViSenze's recognition API).
- * **Applications:** Catalogue Enrichment (Lazada), Intelligent Ad Technologies (Contextual In-Image/In-Video Ads).
- * **Other Vertical Domains:** Home furnishing search, Product search (snap and match), Location identification, Face Recognition, Surveillance.
- * **Social Media (Microblogs):** Many posts have images with no text/unrelated tags. Need to perform MM-based event detection and product/brand tracking.
- * **Visual-based Popular Apps:**
 - **SnapChat:** Temporal Real-Time Messages (10-sec video). Promotes creativity (Drawing, geo-filters). Has a "Discover" feature for news/brands.
 - **Periscope:** Live broadcast from mobile. Interactive, integrates with Twitter.

12.6 New Opportunities

- * **NEXT Center (NUS/Tsinghua):** Center for Extreme Search, aiming to gather and analyze LIVE UGCs related to a city (Live, Big Data, Multifaceted analysis).